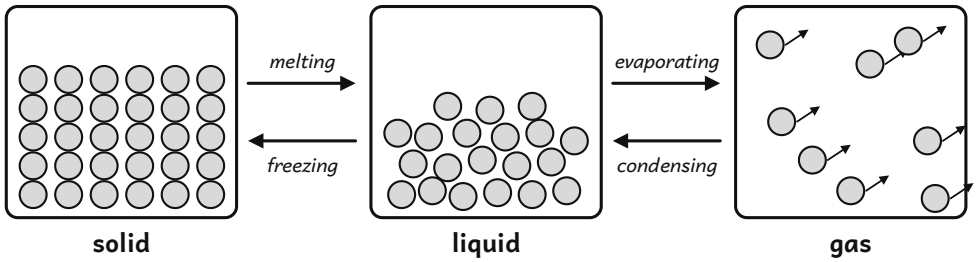
PHYSICAL AND CHEMICAL SCIENCE

Matter: States, Elements and Compounds

Solids, liquids and gases, changes of state, melting and boiling points, pure substances, elements and compounds



In this pack

- **Learn it:** Matter: solids, liquids, gases, elements and compounds ★☆☆
- **Fact check:** States and substances ★☆☆
- **Think it through:** Explaining matter ★★☆
- **Investigate:** The melting race ★★☆
- **Real-world case study:** Ice for the fishing boats ★★★
- **Review and challenge:** Show what you know ★★★

Also covers Cambridge Primary Science Stage 4: Unit 3 Solids, liquids and gases (3.1–3.6).

This pack belongs to: _____ **Class:** _____

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LEARN IT · C1 · MATTER

Matter: solids, liquids, gases, elements and compounds

★☆☆ Grade 4

Name: Class: Date:

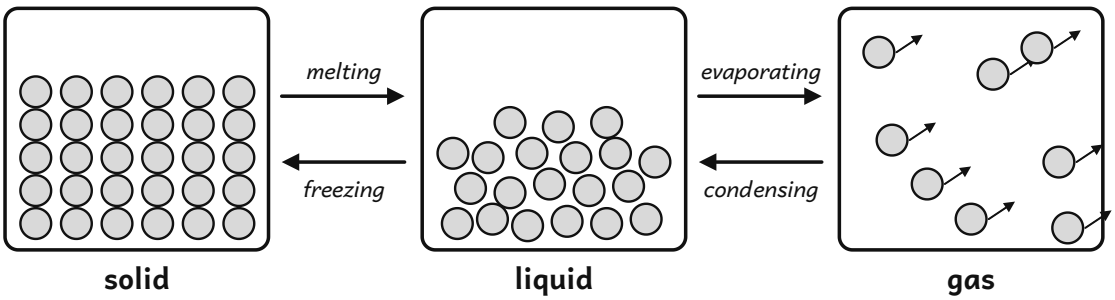
Learning goals

- Describe solids, liquids and gases and model their particles.
- Explain melting, freezing and boiling, and melting and boiling points.
- Tell the difference between pure substances and mixtures, and between elements and compounds.

Your pencil, the water in your bottle, and the air you breathe are all made of **matter**. Matter is anything that has **mass** (it weighs something) and takes up **space**. Even air is matter: blow up a balloon and you can see it take up space!

Three states of matter

Matter is made of particles far too small to see. How the particles are arranged gives each **state** its properties.



Particles in a solid, a liquid and a gas.

State	Particles	Properties	Examples
solid	packed tightly in rows; only vibrate	keeps its shape; cannot be poured or squashed	ice, wood, stone, a spoon
liquid	close together but can slide past each other	flows; takes the shape of its container; keeps its volume	water, milk, oil, juice
gas	far apart, moving fast in all directions	spreads out to fill any space; can be squashed	air, steam, the gas in a cylinder

Remember

Some things are tricky. Sand pours, but each grain is a tiny **solid**. Toothpaste flows when squeezed, but it is a mixture.

Changing state

- **Melting**: a solid becomes a liquid when it is heated (ice → water).
- **Freezing**: a liquid becomes a solid when it is cooled (water → ice).
- **Boiling** and **evaporating**: a liquid becomes a gas (water → water vapour).
- **Condensing**: a gas becomes a liquid when it is cooled (steam → water drops on a lid).

The temperature at which a solid melts is its **melting point**. The temperature at which a liquid boils is its **boiling point**. Different materials have different melting points:

Material	Melting point (about)
ice	0 °C
butter	32 °C
chocolate	34 °C
candle wax	60 °C
iron	1538 °C

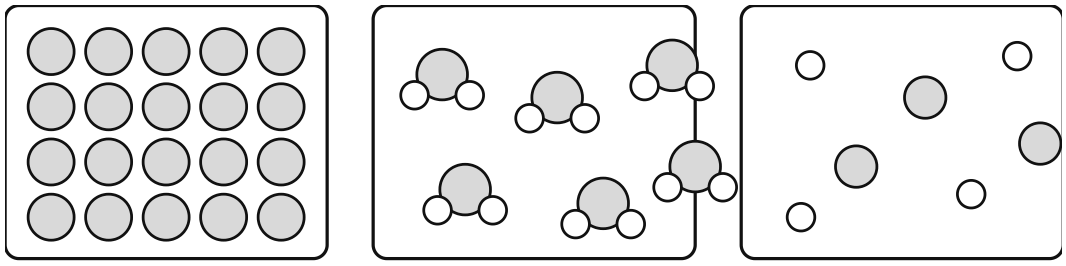
Pure water **freezes at 0 °C** and **boils at 100 °C**. While ice is melting, its temperature stays at 0 °C until all the ice has melted.

Pure substances and mixtures

A **pure substance** contains only one kind of material: pure water, salt, sugar, pure gold. A **mixture** contains two or more substances mixed together: air (many gases), sea water (water and salt), soil, cha with milk and sugar.

Elements and compounds

Pure substances can be **elements** or **compounds**. An **element** is made of only **one kind of atom**: oxygen, iron, gold, carbon. There are about 118 elements. A **compound** is made of **two or more kinds of atom joined together**: water (hydrogen and oxygen), salt (sodium and chlorine), carbon dioxide.



An element, a compound and a mixture.

Using them every day

- Hospitals use pure **saline** (salt and pure water) in drips.
- Gold jewellery is usually a **mixture** of gold with a little copper, which makes it harder.
- **Oral rehydration salt (ORS)** is a carefully measured mixture that helps children with diarrhoea.
- We breathe **air**, a mixture. Our bodies use the **oxygen** in it.

Bangladesh connection

Oral rehydration solution was developed and tested in Bangladesh and India in the 1960s and 70s, including at icddr,b in Dhaka. It has saved millions of children's lives around the world.

Key words

matter	anything that has mass and takes up space
solid	keeps its shape
liquid	flows and takes the shape of its container
gas	spreads out to fill any space
melting point	the temperature at which a solid melts
boiling point	the temperature at which a liquid boils
element	a substance made of only one kind of atom
compound	two or more kinds of atom joined together
mixture	substances mixed but not joined

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LEARN · PLAY · WONDER

FACT CHECK · C1 · MATTER

States and substances

★☆☆
Grade 4

Name: Class: Date:

Learning goals

- Sort solids, liquids and gases.
- Name changes of state.
- Sort elements, compounds and mixtures.

1 Sort the materials. [9]

SORT THESE

rice grain **milk** **ice** **stone** **cooking gas** **air in a balloon** **steam** **rain water** **mustard oil**

Solid	Liquid	Gas

2 Name the change of state. [4]

What happens	Change of state
ice cream drips on a hot day	
water is put in a freezer and becomes ice	
a puddle disappears in the sun	
drops form on a cold glass of water	

3 Draw the particles in each state. [3]

solid

liquid

gas

4 **Sort the substances.** [9]

SORT THESE

gold **water** **oxygen** **iron** **soil** **sea water** **air** **carbon dioxide** **salt**

Element	Compound	Mixture

Explaining matter

★★☆
Grade 4

Name: Class: Date:

Learning goals

- Explain properties using particles.
- Use melting points.

1 **Use the particle model to explain why:** [4]

a) a liquid takes the shape of its container.

b) you can squash a gas but not a solid.

2 **Chocolate melts at 34 °C and candle wax melts at 60 °C. On a 38 °C summer day, which one melts in your bag? Why?** [2]

3 **Rumi says, "Salt is an element because it is pure." Is she right? Explain.** [2]

4 **Why is it useful that iron has a very high melting point?** [2]

INVESTIGATE · C1 · MATTER

The melting race

★★☆
Grade 4

Name: Class: Date:

Learning goals

- Plan a fair test.
- Time and record results.
- Link results to melting points.

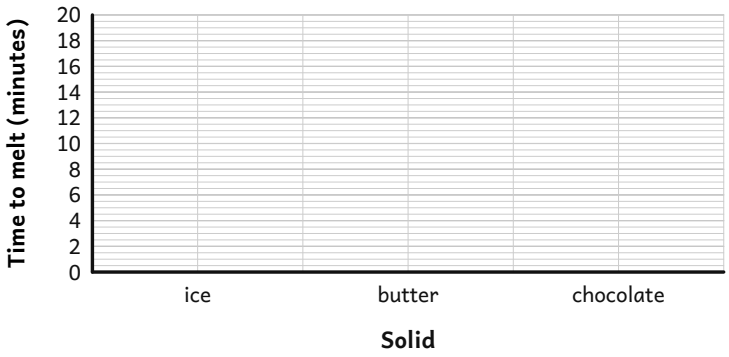
Class 4B put a 1 cm cube of four solids on four identical metal lids in the same sunny place, and timed how long each took to melt completely.

1 **Complete the plan.** [4]

Change	
Measure	
Keep the same	
Prediction	

2 **Draw a bar chart of the results.** [3]

Solid	ice	butter	chocolate	candle wax
Time to melt (minutes)	6	14	18	did not melt



3 **Why did the candle wax not melt?** [1]

.....

.....

4 **Describe the pattern between melting point and time to melt.** [2]

.....

.....

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REAL-WORLD CASE STUDY · C1 · MATTER

Ice for the fishing boats

★★★ Grade 4

Name: **Class:** **Date:**

Learning goals

- Apply changes of state to a real job.
- Use numbers.
- Think about energy and waste.

Science · Economics · Maths · Health

Before a fishing trawler leaves Chattogram or Cox's Bazar for the Bay of Bengal, workers load it with tonnes of **crushed ice**. A trip can last a week or more. The fishers pack each catch of hilsa and other fish in ice so it stays fresh until they return.

The ice is made in **ice factories** near the harbour. Machines cool water below **0 °C** until it **freezes** into big blocks, which are crushed before loading. As the ice slowly **melts**, it takes heat from the fish and keeps them cold. Without ice, fish start to go bad in a few hours in the heat.

Based on information about fish preservation in Bangladesh's fishing industry. Numbers below are examples.

1 **Which changes of state happen in the story?** [2]

Where	Change of state
in the ice factory	
in the boat's fish hold	

2 **Why is crushed ice better than one big block for keeping fish cold?** [2]

3 **Maths. A boat needs 2 kg of ice for every 1 kg of fish. It hopes to catch 500 kg. How much ice does it need?** [2]

4 **Making ice uses a lot of electricity. Suggest one way to waste less ice.** [2]

5 **Why is it important for people's health that fish are kept cold?** [1]

Show what you know

★★★
Grade 4

Name: Class: Date:

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 Find the key words.

W	R	N	B	I	Y	Y	W	S	L
L	I	Q	U	I	D	R	U	O	M
C	N	N	P	F	B	P	W	L	J
M	A	T	T	E	R	O	A	I	S
U	V	M	L	G	C	E	I	D	B
E	L	E	M	E	N	T	E	L	O
V	R	L	S	M	J	K	O	Z	K
U	I	T	S	S	A	F	G	I	E
M	I	X	T	U	R	E	L	A	R
W	M	V	K	W	U	C	G	U	S

boil element freeze gas liquid matter melt mixture solid

2 At what temperature does pure water boil? [1]

- ◯ 0 °C
- ◯ 50 °C
- ◯ 100 °C
- ◯ 34 °C

3 Which is a compound? [1]

- ◯ oxygen
- ◯ water
- ◯ gold
- ◯ air

4 Draw a diagram to show what happens to the particles when ice melts. [3]

Ice melting

5 **Challenge. A glass of iced water gets wet on the outside. Where does the water come from? Explain using states of matter.** [2]

.....

.....

.....

How am I doing? Colour or tick a circle.

I can...	Not yet	Nearly	Got it!
describe solids, liquids and gases using particles	○	○	○
name changes of state	○	○	○
use melting and boiling points	○	○	○
tell pure substances from mixtures	○	○	○
tell elements from compounds	○	○	○
explain how ice keeps fish fresh	○	○	○